Anthony Lu

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Education

New York University **New York, NY**
B.A. Computer and Data Science, Minor in Mathematics - **GPA 3.76/4.00** Expected May 2027
· **Coursework:** Deep Learning & LLMs, Machine Learning, Programming Languages, Parallel Computing, Computer Systems, Algorithms, Data Structures, Linear Algebra, Probability & Statistics, Calculus III, Differential Equations.

Experience

InterSystems **Boston, MA**
Software Engineer Intern May 2026 - August 2026
· Developed an agentic AI assistant (**ObjectScript**, **Python**) for generating data transformations within InterSystems Interoperability, accelerating development time by **16x** over manual authoring.
· Built validation tools (**Rust**) and context framework, improving **accuracy by 15%+** and **token efficiency by 35%+**.
· Engineered an automated synthetic data generation and **LLM evaluation** harness across multiple client use cases, evaluating performance metrics and surfacing domain-specific model failure modes.

DermImpact **New York, NY**
Founding Engineer February 2026 - May 2026
· Built and deployed a full-stack platform (**Next.js**, **TypeScript**) for AI-assisted dermatology notes and image analysis, validated in a live pilot with a **30+ physician dermatology clinic**.
· Architected privacy-focused data pipelines with RLS, encrypted image storage, and PostHog usage analytics.

NYU Biokind Analytics **New York, NY**
Data Engineer January 2026 - May 2026
· Built FastAPI-based **data pipelines** processing **35k+** nonprofit donor records, automating database maintenance and reducing manual processing time by **60%** while increasing platform engagement by **20%**.

Pulse Foundry **New York, NY**
Software Engineer Intern December 2025 - February 2026
· Built and deployed an automated content recommendation pipeline (**FastAPI**, **OpenAI**) with **concurrent ingestion**, **semantic ranking**, and **scheduled workers**, serving **3,000+ subscribers** in production.
· Developed an internal **React** dashboard enabling editors to review, filter, and configure pipeline parameters in real time.

Projects

hLLM | *C++, Python, CUDA, MLX, gRPC* August 2026 - Present
· Developed a heterogeneous **LLM inference** runtime that partitions models across Apple Silicon and NVIDIA GPUs, executing **distributed prefill** and autoregressive decoding over gRPC.
· Implemented native **CUDA** and **MLX** backends with Safetensors weight loading and cross-device activation streaming.
· Built hardware-aware profiling and placement to optimize model partitioning across memory, compute, and network costs.

HyperZero | *Python, PyTorch, FastAPI, React, Three.js* May 2026 - June 2026
· Built a **reinforcement learning agent** for multidimensional Connect-K combining Monte Carlo Tree Search, self-generated data, and policy-value neural networks.
· Achieved **87% mean win rate** against heuristic and tree-search baselines across multiple board configurations.
· Developed a training pipeline optimized for high-throughput with **batched inference** and **parallelized simulations**.
· Deployed a real-time 3D game interface with WebSocket-based agent API and live move visualizations.

Neural-ViT | *Python, PyTorch, Vertex AI, BigQuery, Google Cloud Storage* December 2025 - January 2026
· Implemented a **3D Vision Transformer** for genotype classification on multi-trial neural spectrograms, achieving **0.96 AUC** (+9% over baseline models), trained on NVIDIA A100s via Google Cloud Vertex AI.
· Built an end-to-end ML pipeline processing **77k+ trials** with preprocessing, experiment tracking, and reproducible training.
· Reduced training time by **15%** using mixed-precision training and optimized data loading, with LR scheduling and dropout for stable generalization.

Skills

Languages: Python, C++, C, JavaScript, TypeScript, Java, SQL, Rust, Go, HTML, CSS
ML/Data: PyTorch, scikit-learn, Weights & Biases, Pandas, NumPy, Hugging Face Transformers
Frameworks: React, Angular, Node.js, Next.js, FastAPI, Express.js, PostgreSQL, SQLite
Infrastructure: GCP, AWS, Docker, Git, Linux, CUDA